

SamPave Demo Workflow Guide — NH-99 Project

CRITICAL SAFETY DISCLAIMER

IMPORTANT:

SamPave is a professional decision-support engineering tool. All empirical design projections, thickness suggestions, material specifications, and mechanistic calculations generated by this software are for engineering screening and decision-support only. Final pavement designs, thickness profiles, and mix configurations must be independently verified, reviewed, and signed off by a licensed and qualified Professional Pavement Engineer before field execution, bidding, or construction.

1. Project Background

This demo workflow walks through the pavement design of the **National Highway NH-99 Corridor** (4-lane divided carriageway, subgrade soaked CBR of 4.0%). The goal is to design a structurally sound flexible pavement with stabilized layers to optimize material usage and minimize bituminous cover thickness.

2. Step-by-Step Walkthrough

2.1 Project Creation

1. Open the Project Hub and click **Create New Project**.
2. Set:
 - **Work Name:** NH-99 4-Lane Divided Carriageway
 - **Client:** NHAH Regional Office
 - **Consultant Name:** Pavement Consultants Group
 - **Submitted By:** QA Lead Auditor
 - **Mix Type:** DBM-II
 - **Binder Grade:** VG-30
3. Click **Create Project**.

2.2 Traffic count and MSA Calculation

1. Go to the **Traffic Analysis** tab.
2. Input raw traffic survey data:
 - **Initial Commercial Traffic (CVPD):** 2500 (derived from 7-day volume counts)
 - **Annual Traffic Growth Rate (%):** 6.5

- **Design Life (years):** 15
- **Vehicle Damage Factor (VDF):** 4.5 (from axle-load survey)
- **Lane Distribution Factor (LDF):** 0.75 (typical for 4-lane divided carriageways, per IRC:37 cl. 4.4)

3. Click **Compute Traffic**.

• *Result:* Cumulative traffic calculated = **66.25 MSA**.

• *Growth Factor:* **23.716**.

4. Click **Save & Continue**.

2.3 Subgrade CBR and Modulus calculation

1. Go to the **Subgrade Properties** tab.

2. Input laboratory soaked CBR values:

• **Subgrade Soaked CBR (%):** 4.0

3. Click **Calculate Modulus**.

• *Result:* Modulus (\$M_r\$) = **40.0 MPa** (calculated as \$10 times 4.0%\$, since CBR $\leq 5\%$).

4. Click **Save**.

2.4 Catalogue design lookup

1. Go to the **Catalogue Design** tab.

2. Click **Lookup Catalogue Composition**.

• *Result:* The catalog query finds the closest match for **CBR 3-5%, MSA 50-100**:

• Bituminous Concrete (BC): 40 mm

• Dense Bituminous Macadam (DBM): 170 mm

• Wet Mix Macadam (WMM): 250 mm

• Granular Sub-base (GSB): 300 mm

3. Click **Adopt Suggestion** to load this into the Structural design workspace.

2.5 Stabilized Design Wizard (CTB/CTS Option)

1. Go to the **Stabilized Design** tab.

2. Input stabilized layer target values:

• **Cement-Treated Base (CTB) Thickness:** 120 mm

• **CTB Modulus:** 5000 MPa

• **CTB 28-day UCS Strength:** 4.5 MPa

- **Cement-Treated Sub-base (CTS) Thickness:** 100 mm
- **CTS Modulus:** 3000 MPa
- **Granular Sub-base (GSB) Thickness:** 150 mm
- **Bituminous Cover Thickness:** 100 mm

3. Click **Calculate Stabilized Composition**.

- **Result:** Nominal bituminous thickness savings = **110 mm** (a 52% saving in expensive asphalt binders).
- **Comparison mode:** Decision Support Mode (unless **IITPAVE.exe** is configured).

2.6 Engineering Intelligence Screen

1. Navigate to the **Structural Pavement Design** tab.
2. Review the **Engineering Screening Score** and warnings on the right side:
 - **Stiff Layer over Weak Support:** CTB is placed directly over GSB. Since GSB modulus is 150 MPa (< 500 MPa), this triggers a warning: *"Very stiff stabilized layer 'CTB' is placed directly over weak support. Potential risk of cracking due to stiffness mismatch."*
 - **Stiffness Inversion:** Reviewed; layers follow structural tier order.
 - **Screening Score:** Displays **65/100** (Needs engineering review).
3. Review findings and note in the design diary that shrinkage interlayers will be specified.

2.7 IITPAVE Mechanistic Verification (If exe configured)

1. If **IITPAVE.exe** is configured in the **IITPAVE Manager**, click **Run Mechanistic Validation**.
2. The engine runs in the background and parses the strain outputs:
 - Fatigue tensile strain (ϵ_t) at bottom of bituminous layer.
 - Vertical compressive strain (ϵ_v) at top of subgrade.
3. The system confirms if fatigue and rutting lives exceed design requirements.

2.8 Review checklist and Project Locking

1. Navigate to the **Submission Center**.
2. Complete the **Consultant Checklist**:
 - Check: *Traffic counts verified*
 - Check: *Subgrade soil soaked CBR test verified*
 - Check: *Layer modular ratios reviewed*
3. Set review status to **Reviewed**.
4. Click **Lock Project** to secure the design parameters from modification.

2.9 DPR Export and Submission ZIP Package

1. Click **Export DPR**. Choose file name `NH_99_Detailed_Project_Report.docx` and click **Save**.
2. Click **Export Package (.zip)**. Choose file name `NH_99_Submission_Package.zip` and click **Save**.
3. The ZIP package compiles:
 - `NH_99_Detailed_Project_Report.docx` (Word DPR)
 - `ProjectInputs.json` (Database snapshot)
 - `README_Disclaimer.txt` (Disclaimer text)
 - `ApprovalSheet.txt` (Checks sheet)